

Accompanying Notes

Codebook procedure

There are several studies by the original authors of the IGT describing different versions of the IGT. The codebook was designed to capture the original IGT (Bechara et al., 1994) as well as its computerized version (Bechara et al., 1999).

However, the clinical version of the IGT (Bechara 2007, 2016), which builds on Bechara et al., 2000 (*Characterization of the decision-making deficit of patients with ventromedial prefrontal cortex lesions*), differs from the layout of the codebook in a significant way: In these studies, the pay-out schedule changes over the course of the task (i.e., magnitude and frequency of punishments as well as magnitude of immediate rewards can change). So, whenever a study clearly follows Bechara et al. (2000), Bechara (2007), Bechara (2016), or describes another pay-out schedule which changes as the task progresses, the codebook can't capture the procedure adequately.

To deal with this, the following codes were used:

VSDot = "variable schedule, determinable, original task"

The changes regarding the pay-out schedule are determinable and are programmed according to either Bechara et al.'s (2000) A'B'C'D'-variant or Bechara (2007, 2016).

VSDev = "variable schedule, determinable, E'F'G'H' variant"

The changes regarding the pay-out schedule are determinable and are programmed according to Bechara et al.'s (2000) E'F'G'H'-variant.

VSI = "variable schedule, indeterminable"

The changes regarding the pay-out schedule are (fully or partially) indeterminable. Information provided is captured in the notes_procedure column.

Codebook scoring

u = "unique"

A placeholder for a unique score id.

u_ind = "unique, indeterminable"

A placeholder for a unique score id. At least one column is indeterminable.

MSI = "multiple scores indeterminable"

The paper uses more than one score to measure IGT performance, but the scores are indeterminable. Information provided is captured in the notes_scoring column.

PLACEHOLDER

The placeholder indicates instances where I couldn't decide on how (or if) to code a particular score. This inability was not (necessarily) due to ambiguity in the text but lack of knowledge from my side regarding the statistical models used in the paper. For more information see the respective `notes_scoring` columns.

EV, TDRL, and VPP models

To analyse IGT performance trial by trial, several studies used mathematical models. The models used are:

- EV model = "Expectancy Valence model" (Busemeyer & Stout, 2002)
- TDRL model = "Temporal Difference Reinforcement Learning model" (Christakou et al., 2013)
- VPP = "Value-Plus-Perseveration model" (Worthy, Pang, & Byrne, 2013)